

POWER ELECTRONICS

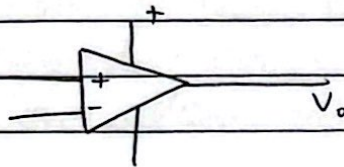
Date: 20/01/2025

Sun Mon Tue Wed Thu Fri Sat

DAY-1

AC $\rightarrow$ DC	Rectification
DC $\rightarrow$ AC	Inverter
DC $\rightarrow$ DC	choppers
AC $\rightarrow$ AC	Cyclon converters (changes Freq. ^(speed))

Operational Amplifier :



Negative Feedback :

Output must be fed back to the inverting input.

Feedback ratio is β , where $0 \leq \beta \leq 1$.

closed loop gain $A_{cl} \approx \frac{1}{\beta}$

$$e_{inv} = V_f = \beta e_{out}$$

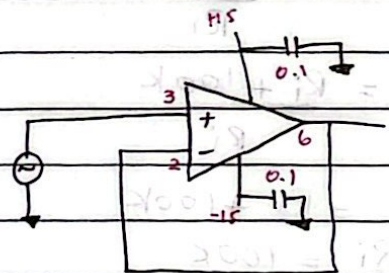
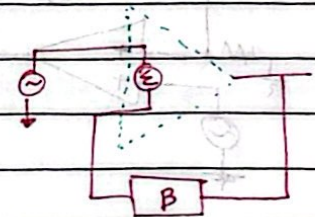
$$e_{ni} = e_{in}$$

$$V_d = e_{ni} - e_{inv}$$

$$0 = e_{in} - \beta e_{out}$$

$$e_{in} = \beta e_{out}$$

$$e_{out} = e_{in}$$



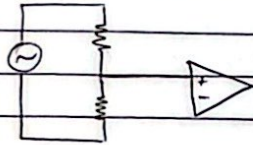
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Example 1

DAY-1

$$R = \frac{1}{\frac{1}{500} + \frac{1}{1000}} = 333.33 \Omega$$

$$V_o = \frac{333}{1333} \times 1 \Rightarrow 0.25V$$

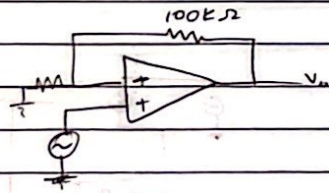


(incomplete)

Non-inverting Amplifier :

$$A_{cl} = 1 + \frac{R_F}{R_i}$$

Example 2



$$100 = 1 + \frac{100k}{R_i}$$

$$100 = \frac{R_i + 100k}{R_i}$$

$$100R_i = R_i + 100k$$

$$99R_i = 100k$$

$$R_i = \frac{100k}{99} \Rightarrow 1010 \Omega$$

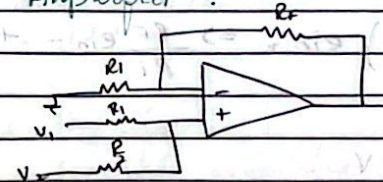
Inverting Amplifier :

$$A_{cl} = -\frac{R_F}{R_i}$$

Universal Analysis Techniques:

- Find voltage on non-inverting input pin.
- Find voltage on inverting input pin
- Calculate current through each of the input resistor
- Calculate current through R_F using KCL
- Calculate voltage drop across R_F
- Calculate output voltage using KVL

Summer Amplifier :



$$V_o = \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_2}{R_1 + R_2}\right) V_1 \quad (\text{Finding for } V_1, \text{ putting } V_2 = 0)$$

$$+ \left(1 + \frac{R_F}{R_i}\right) \left(\frac{R_1}{R_1 + R_2}\right) V_2 \quad (\text{Finding for } V_2, \text{ putting } V_1 = 0)$$

DAY-2

Example 3:

$$V_{NT} = 0V$$

Date:.....

Sun Mon Tue Wed Thu Fri Sat

$$\begin{aligned}
 & \text{ii) } V_{INV} = 0V \\
 & \text{iv) } I_{R1} = \frac{1-0}{5.6K} = 179 \mu A, I_{R2} = \frac{15-0}{470K} = 32 \mu A \\
 & \text{iv) } 179 \mu + 32 \mu = 211 \mu A = I_{RF} \quad (\rightarrow) \\
 & \text{v) } V_{RF} = (211 \mu)(10K) \Rightarrow 2.11m \text{ or } 2.11V \\
 & \text{vi) } E_{out} = -2.11 + 0 \Rightarrow -2.11V \\
 & E_{out} = (-2.11)(-1) \Rightarrow 2.11V
 \end{aligned}$$

Difference Amplifier:

$$e_{out} = \frac{R_F}{R_i} (e_{in+} - e_{in-}) \text{ so } \frac{-R_F}{R_i} e_{in-} + \left(\frac{1+R_F}{R_i} \right) \left(\frac{R_F}{R_F+R_i} \right) e_{in+}$$

if same,

$$= -\frac{R_F}{R_i} e_{in-} + \left(\frac{R_i+R_F}{R_i} \right) \left(\frac{R_F}{R_i+R_F} \right) e_{in+} \Rightarrow \frac{-R_F}{R_i} e_{in-} + \frac{R_F}{R_i} e_{in+}$$

Example 4:

$$1) V_{NI} = 105K \quad (5\% = 4.955V)$$

$$2) V_{INV} = 4.955V$$

$$3) I_{R1} = \frac{5-4.955}{1050} \Rightarrow 42.86 \mu A$$

$$4) I_{RF} = 42.86 \mu A$$

$$5) V_{RF} = (42.86 \mu)(95K) \Rightarrow 4.056$$

$$6) E_{out} = 4.955 - 4.071 = 885mV$$

Instrumentation Amp

Example 5:

Date:.....

Sun Mon Tue Wed Thu Fri Sat

V1	E-VATU 2
i) $V_{NI} = 2.45V$	i) $V_{NI} = 2.55V$
ii) $V_{INV} = 2.45V$	ii) $V_{INV} = 2.55V$
iii) $I_{R1} = \frac{2.45-0}{9K}$	iii) $I_{R1} = \frac{2.55-2.22}{1K}$
$= 272 \mu A (\leftarrow)$	$= 328 \mu A$
iv) $I_{R2} = \frac{2.55-2.45}{200\Omega}$	iv) $I_{R2} = \frac{328 + 500}{828 \mu A}$
$= 500 \mu A$	
v) $I_{RF} = 272 + 500 = 772 \mu A$	v) $V_{RF} = (828 \mu A)(9K)$
$= 228 \mu A$	$= 7.452V$
vi) $V_{RF} = (228 \mu)(1K)$	vi) $E_{out} = 2.55 + 7.45$
$= 0.228V \text{ or } 228mV$	$A_4 = 10V$
vii) $E' = 2.45 - 0.228 = 2.222V$	
$E' = 2.222V$	

Composite Amplifier:

Composes several amplifiers

Example 6

$$1) V_{NI} = 0V$$

$$2) V_{INV} = 0V$$

$$iii) I_{R1} = \frac{2.56}{10K} \times 128 \Rightarrow 128 \mu A$$

$$iv) I_{RF} = 128 \mu A$$

$$v) V_{RF} = (128 \mu)(39.2K) \Rightarrow 5.02$$

$$vi) E_{out} = 5.02V$$

$$E_{out} = 5.02 - 1.2 \Rightarrow 3.78V$$

Date: 27/01/2025

Sun Mon Tue Wed Thu Fri Sat

DAY-3

Active feedback composite amp.:

Amp. as part of negative feedback loop

Example 7:

U1

U2

1) $V_{inv} = 4V$

2) $V_{inv} = 4V$

3) $I_{R_{ref}} = 4$

$10K + 1.1K$

$= 360.36 \mu A$

1) $V_{inv} = (360.4 \mu)(1.1K)$

$V_{inv} = 396.4 mV$

2) $V_{non} = 396.4 mV$

3) $I_{R_{ref}} = 396.4 mV$

$0.1K$

$V_{out} = 3.964A$

$V_{emitter} = 3.964(0.1 + 2)$

$E_{out} = 8.32 + 1.8$

$= 10.1V$

Suspended Supply Amplifier
Supply is varying

Example 8:

$X_{min} =$

$V_{inv} = \frac{6.8}{6.8 + 1} (E_{out})$

$= 0.87 E_{out}$

$V_{NI} = 0.87 E_{out}$

$I_{R_{ref}} = \frac{6 - 0.87 E_{out}}{10}$

$E_{out} - 6 = 1 \quad E_{out} = 6$

$1.8 + 10 \quad 1.8 + 10 \quad 1.8 + 10$

$= 847 \times 10^{-6} E_{out} - 508.4 \times 10^{-6}$

$V_{NI} = 6 + I_{R_{ref}} R_a$

$= 6 + (847 \times 10^{-6} E_{out} - 508.4 \times 10^{-6}) 10K$

$= 6 + 8.47 \times 10^{-3} E_{out} - 5.084$

$= 6 + 0.847 E_{out} - 5.084$

$0.87 E_{out} = 0.916 + 0.847 E_{out}$

For supply, we will give ans in $\pm$ $E_{out} = 39.82V$

Date:

Sun Mon Tue Wed Thu Fri Sat

POWER PARAMETER CALCULATIONS

→ Average Value

→ RMS

→ Power Delivered

DC Waveform:

→ Const. value

→ Remains equal to peak value

$V(t) = V_p$

Sinusoidal Waveform

$V(\theta) = V_p \sin \theta$ or $V(t) = V_p \sin(\omega t)$

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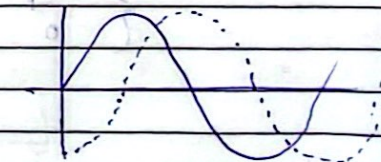
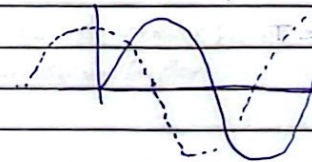
$V(\theta) = -V_p \sin \theta$

$V(\theta) = -V_p \sin(\theta + \pi)$ or $V_p \sin(\theta - \pi)$

Effect of capacitor & Inductor:

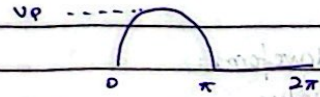
$i(\theta) = I_p \sin(\theta + \frac{\pi}{2})$

$i(\theta) = I_p \sin(\theta - \frac{\pi}{2})$

Capacitor
(lead) by 90° Inductor
(lag) by 90° 

Half Wave Rectifier :

$$V(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

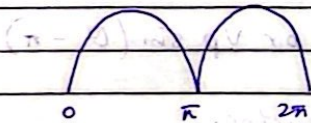
AC $\rightarrow$ DC conversion

$$V(\theta) = \begin{cases} 0 & 0 \leq \theta < \pi \\ +V_p \sin \theta & \pi \leq \theta \leq 2\pi \end{cases}$$

+ve because it is down. If flipped, it will have -ve!

Full Wave RectifierAC $\rightarrow$ DC conversion

$$V(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ -V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$$



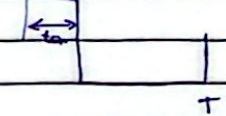
$$V(\theta) = \begin{cases} -V_p \sin \theta & 0 \leq \theta < \pi \\ V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$$

Rectangular Wave:

$$\begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

DAY-4Rectangular Wave:

$$D = \frac{t_{on}}{T}$$

Triangular Wave:

$$y = mx + c$$

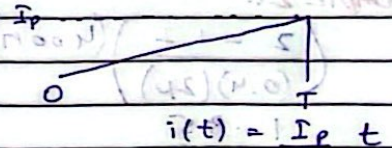
it is a ramp signal

$$f(x) = mx + c$$

$$m = \text{slope} = \frac{\text{Rise}}{\text{Run}}$$

$$i(t) = \frac{I_p}{T} t + 0$$

$$= \frac{I_p t}{T} \quad 0 \leq t < T$$



Example 1:-

$$\begin{aligned} & 3 \quad t \\ & 6.66 \mu \\ & 0.45 \mu t \\ & (0.45 \mu)(75 \mu) \\ & = 2.25 \end{aligned}$$

For 10 us

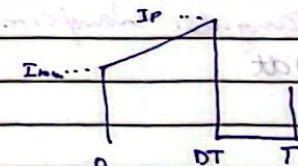
$$\Rightarrow 10 \mu s = 6.67 \mu$$

= ANS

then put in equation

For 15 us

$$t = 15 \mu s$$

Trapezoid Waveform: $\rightarrow$ Due to inductor

Date:.....

Sun Mon Tue Wed Thu Fri Sat

N-VAD

$$i(t) = \begin{cases} 0 & DT \leq t < T \\ \left(\frac{I_p - I_{min}}{DT} \right) t + I_{min} & 0 \leq t < DT \end{cases}$$

$$y = mx + b$$

initial value of line = I_{min}

$$\left(\frac{I_p - I_{min}}{DT} \right) t + I_{min}$$

Example 2:

$$\left(\frac{2 - 1.7}{(0.4)(2\mu)} \right) (400n) + 1.7 \rightarrow \text{For } 400n \text{ s}$$

$$= 1.85$$

$$\left(\frac{2 - 1.7}{(0.4)(2\mu)} \right) (1\mu) + 1.7 \rightarrow \text{For } 1\mu \text{ s}$$

$$= 0$$

$$\therefore I_{min} = I_p - \text{ripple}$$

$$i(t) = \begin{cases} \frac{2 - 1.7}{800n} t + 1.7 & 0 \leq t < 800n \text{ s} \\ 0 & 800 \leq t < 2\mu \text{ s} \end{cases}$$

AVERAGE VALUE:

Average value of DC waveform:

→ known as dc value

→ Equal to peak value of dc signal

$$V_{dc} = V_{avg} = V_p$$

Average value of rectangular waveform:

$$V_{avg} = \frac{1}{T} \int_0^T v(t) dt$$

$$v(t) = V_p$$

Date:.....

Sun Mon Tue Wed Thu Fri Sat

$$V_{avg} = \frac{1}{T} \int_0^T V_p dt$$

$$= \frac{V_p}{T} t \Big|_0^T$$

$$= V_p$$

For rectangular wave

$$v(t) = \begin{cases} V_p & 0 \leq t \leq DT \\ 0 & DT \leq t < T \end{cases}$$

$$V_{avg} = \frac{1}{T} \left[\int_0^{DT} V_p dt + \int_{DT}^T 0 dt \right]$$

$$= \frac{1}{T} [V_p t]_0^{DT}$$

$$= V_p \frac{DT}{T}$$

$$V_{avg} = D V_p$$

For sine wave or cos wave:

Average value of a sinusoid is 0

For half wave rectifier:

$$V_{avg} = \frac{V_p}{\pi}$$

For full wave rectifier:

$$V_{avg} = \frac{2V_p}{\pi}$$

DAY-5

DC signal.

$$V_{avg} = V_p$$

Rectangular signal

$$V_{avg} = DV_p$$

Sinusoidal

$$V_{avg} = 0$$

Half-wave

$$V_{avg} = \frac{V_p}{\pi}$$

Full-wave:

$$V_{avg} = \frac{2V_p}{\pi}$$

Triangular Wave:

$$i(t) = \frac{I_p}{T} t$$

$$I_{avg} = \frac{1}{T} \int_0^T \frac{I_p}{T} t dt$$

$$= \frac{I_p}{T^2} \int_0^T t dt$$

$$= \frac{I_p}{2}$$

Trapezoid wave:

$$\frac{1}{T} \int_0^{DT} (I_p - I_{min}) t + I_{min} dt$$

$$= D \left[\frac{I_p - I_{min} + 2 I_{min}}{2} \right]$$

$$= \frac{(I_p + I_{min}) D}{2}$$

Example:

$$I_p = 3A$$

$$I_m = 3 - 400m$$

$$= 0.35 \left[3 - 2.6 + 2(2.6) \right]$$

$$I_{avg} = 0.98 A$$

Root mean square value:

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T V^2(t) dt}$$

DC signal.

$$V(t) = V_p = V_{avg} = V_{rms}$$

Rectangular signal.

$$V_{rms} = V_p D$$

Sin wave: $V(t) = V_p \sin \theta$ $0 \leq \theta < 2\pi$

$$V_{rms} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} V_p^2 \sin^2 \theta d\theta}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{2\pi}}$$

$$= \sqrt{\frac{V_p^2}{2\pi} \left[\frac{1}{2} (2\pi - 0) - \frac{1}{4} (\sin 4\pi - \sin 0) \right]}$$

$$= \sqrt{\frac{V_p^2}{2\pi}}$$

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

Half Wave

$$V_{rms} = \frac{V_p}{2}$$

DAY-6

$$i(t) = \frac{I_p}{T} t \Rightarrow I_{rms} = \sqrt{\frac{1}{T} \int_0^T \left(\frac{I_p}{T} t \right)^2 dt}$$

$$= \sqrt{\frac{1}{T} \left(\frac{I_p^2}{T^2} \int_0^T t^2 dt \right)} \Rightarrow \sqrt{\frac{I_p^2 T^3}{T^3 \cdot 3}}$$

$$= \frac{I_p}{\sqrt{3}}$$

Trapezoidal Wave:

$$I_{rms} = \sqrt{\frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{min}}{DT} t + I_{min} \right)^2 dt}$$

$$= \sqrt{\frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{min}}{DT} t + I_{min} \right)^2 dt}$$

$$= \sqrt{\frac{1}{T} \left[\frac{(I_p - I_{min})^2}{DT^2} \int_0^{DT} t^2 dt + 2(I_p - I_{min}) I_{min} \int_0^{DT} t dt + I_{min}^2 \int_0^{DT} dt \right]}$$

$$\begin{aligned}
 &= \frac{1}{T} \left[\frac{(I_p - I_{min})^2}{2} DT + I_{min}^2 DT + I_{min}(I_p - I_{min}) DT \right] \\
 &= \frac{1}{T} \left[\frac{(I_p - I_{min})^2}{2} + I_{min}^2 + I_{min}(I_p - I_{min}) \right] \\
 &= \frac{1}{T} \left[\frac{(I_p - I_{min})^2}{2} + I_{min}^2 + I_{min}I_p - I_{min}^2 \right] \\
 &= \frac{1}{T} \left[\frac{I_p^2 + I_{min}^2 - 2I_p I_{min} + 2I_p I_{min}}{2} \right] \\
 &= \frac{1}{T} \left[\frac{I_p^2 + I_{min}^2 + 2I_p I_{min}}{2} \right] \\
 &= \frac{1}{T} \left[\frac{I_p^2 + I_p I_{min} + I_{min}^2}{2} \right]
 \end{aligned}$$

Power Calculations :-

$p(t) \rightarrow$ Pinse $\rightarrow$ Instantaneous Power

$$p(t) = v(t) \cdot i(t)$$

Average power :-

$$\frac{1}{T} \int_0^T p(t) dt$$

DC:

$$v(t) = V_p$$

$$i(t) = I_p$$

$$P = \frac{1}{T} \int_0^T v(t) i(t) dt$$

$$= \frac{1}{T} \int_0^T V_p I_p dt$$

$$= \frac{V_p I_p}{T} t \Big|_0^T \Rightarrow P = VI$$

Rectangular :

$$P = DV_p I_p$$

Resistive load $\rightarrow V$ & I in phase

Sinusoidal

$$v(t) = V_p \sin \theta$$

$$P = \frac{1}{2\pi} \int_0^{2\pi} V_p \sin \theta I_p \sin \theta d\theta$$

$$= \frac{V_p I_p}{2\pi} \int_0^{2\pi} \sin^2 \theta d\theta$$

$$= \frac{V_p I_p}{2\pi} \int_0^{2\pi} \frac{1 - \cos 2\theta}{2} d\theta$$

$$= \frac{V_p I_p}{4\pi} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^{2\pi}$$

$$= \frac{V_p I_p}{4\pi} (2\pi) \Rightarrow \frac{V_p I_p}{2}$$

$$V_{rms} = \frac{V_p}{\sqrt{2}} \Rightarrow V_p = \sqrt{2} V_{rms}$$

$$P = \sqrt{2} V_{rms} \sqrt{2} I_{rms}$$

$$P = V_{rms} I_{rms}$$

$$P = \frac{1}{2\pi} \int_0^{2\pi} V_p I_p \sin^2 \theta d\theta \quad \text{F.YAO}$$

$$= \frac{V_p I_p}{2\pi} (-\cos \theta) \Big|_0^{2\pi} \rightarrow -(-1-1) = -(-2) \rightarrow +2$$

$$= \frac{V_p I_p}{\pi}$$

Example 2:-

$$V_{NI} = 1V_p$$

$$V_{INV} = 1V_p$$

$$I_{IN} = \frac{1}{1.1K} \Rightarrow 0.909mA$$

$$V_{out} = (0.909m)(10 + 1.1)K = 10.08$$

$$I_{Load} = \frac{10.08}{8} \Rightarrow 1.26A$$

$$P_{out} = \frac{(10.08)(1.26)}{2} \Rightarrow 6.35W$$

7.4

Low Power Audio Amplifier:

Example:

$$P_{load} = 0.6 \times V_{load} \times I_{load} = 7.04W$$

$$P_{transist} = 0.6 \times V_{trans} \times I_{trans} = 0.6 \times (1.71) \times 0.2 \times 1.71 = 0.35W$$

DAY-7

Low Power Audio Amplifier:

$$A_{mid} = \frac{2 \times 15K\Omega // 15K\Omega}{1.35K\Omega + 150\Omega} = 10$$

8-VAR

$$\text{Initial value of } C_{bass} = \frac{1}{2\pi \times 150K\Omega \times f_{bass, boost}} \Rightarrow \frac{1}{2\pi Rf}$$



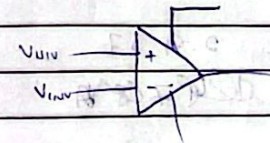
$\approx DC$

$\hookrightarrow$ blocks

$\hookrightarrow$ Acts as open circuit

OPA548 Op. amp:

$$V_{out} = \left(1 + \frac{R_f}{R_i}\right) V_{in,os}$$



$$P = VI$$

$$P_b = V_s I_{load}$$

V_{cos}

$$P_{ic} = P_{supply} - P_{out}$$

Example 3:-

$$P_{load} = \frac{V_{load}^2}{R_{load}}$$

$$= \frac{(5V)^2}{10\Omega}$$

$$I = \frac{5}{10} = 0.5$$

$$P_{supply} = 12 \left(\frac{5}{10} \right) = 6W$$

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

Date:

Sun Mon Tue Wed Thu Fri Sat

DAY-8

Gain bandwidth product is typically 1 MHz

$$GBW = A_o \times f_H$$

$$A_o = \frac{V_o}{V_i}$$

$$SR = \frac{dV_{out}}{dt} \Rightarrow 10 \text{ V}/\mu\text{s} \rightarrow 8 \text{ ohm} \rightarrow$$

Example 2:

Input

$$V_i = 300 \text{ mV}_{rms}$$

$$P_{load} = 35 \text{ W}$$

$$R_{load} = 8 \Omega$$

$$I_L = \frac{35}{8} \Rightarrow 4.375$$

$$P = I_L R$$

$$2.957$$

$$\Sigma = \frac{P}{V_R} \Rightarrow \frac{35}{8} \Rightarrow 4.375 \Rightarrow 2.091 \Rightarrow 2.091$$

$$P = VI$$

$$23.65$$

$$V = \frac{P}{I} \Rightarrow \frac{35}{2.091} \Rightarrow 16.73 \Rightarrow 16.73 \text{ V}$$

$$f = \frac{SR}{2\pi V_p} \Rightarrow \frac{10 \text{ V}/\mu\text{s}}{2\pi(16.73)} \Rightarrow 67.267 \text{ KHz}$$

$$A_o = \frac{V_o}{V_i} = 55.6$$

$$GBW = 55.6 \times 20 \text{ K} \Rightarrow 1.1 \text{ MHz}$$

Date: ONE

Sun Mon Tue Wed Thu Fri Sat

P-VAD

low range amplifiers
50 kHz

Medium range

1-5 MHz

High range

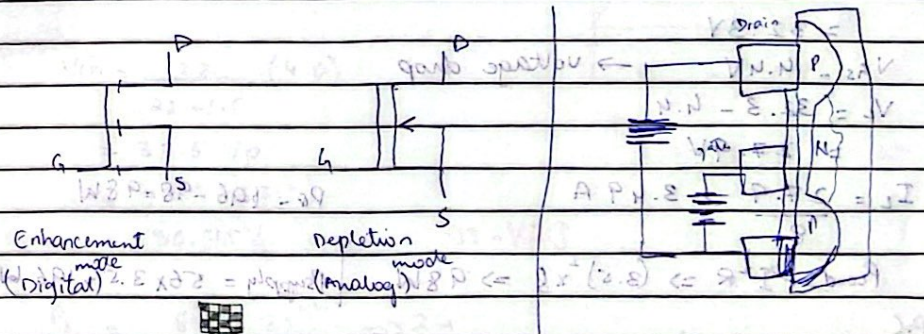
> 5 MHz



till Mid 1

MOSFET & BJT

Enhancement Mode MOSFET



$$V_o = \left(\frac{R_u}{R_3} \right) \left(\frac{1 + 2R_2}{R_F} \right) (V_2 - V_1)$$

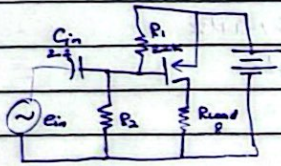
$$\left(\frac{10 \text{ K}}{10 \text{ K}} \right) \left(\frac{1 + 200 \text{ K}}{1 \text{ K}} \right) (10 \text{ mV} - 5 \text{ mV}) \times 1 = 20.21 \text{ V}$$

$$V_o = 1.005 \text{ V}$$

COPY

DAY-9

Example:



DC

$$V_g = \frac{30}{30+22} (56)$$

$$= 32.3V$$

$$V_{gs} = 4.4V \rightarrow \text{voltage drop}$$

$$V_L = 32.3 - 4.4$$

$$= 27.9V$$

$$I_L = \frac{27.9}{8} = 3.49A$$

$$P_o = 196 - 98 = 98W$$

$$P_{load} = I^2 R \Rightarrow (3.5)^2 \times 8 \Rightarrow 98W \quad | \quad P_{supply} = 56 \times 3.5 = 196W$$

AC

$$V_g = e_{in} = 20V \times 0.707 = 14.14V_{rms}$$

$$V_L = (A V_{in}) < 1 = 0.98$$

$$= 0.98(14.14)$$

$$= 13.85V_{rms}$$

$$I_L = \frac{13.85}{8} = 1.73A$$



→

Push Pull Amplifier:

- Transistor is biased off i.e. $V_{gs} = V_{th}$
- When input is 0, transistor is off and that 98W is wasted.

Example:

$$V_{th} = 3.6V$$

$$V_{gs} = 3.6V$$

$$\frac{R_2}{R_2+22} (56) = 3.6$$

$$R_2 = 1.5K\Omega$$

$$V_{in} = \frac{22}{22+1.5} (4.0)$$

$$= 37.5V_p$$

$$V_o = 0.98(37.5)$$

$$= 36.75V_p$$

SI-VAD

$$P_L = 8(1.75)^2 \Rightarrow 24$$

$$P_{sup} = 56(1.75) = 96.8$$

$$P_Q = 96.8 - 24$$

$$= 72.88$$

$$V_L = (0.98)(28.28)$$

$$V_L = 27.7$$

$$I_L = \frac{27.7}{8} \Rightarrow 3.46A$$

$$P_L = (3.46)^2 \times 8 = 96$$

$$= 4V^2 \Rightarrow 42.2W$$

$$P_S = 56 \times 36.75 \Rightarrow 2055W$$

8π

Pull Amplifier:

Same working but processes negative half.

Push-Pull amplifier:

No DC current flows through the load.

No power is dissipated in biasing the transistor.

Same calculations.

Cons of class B Amp:

As

$$I_o \propto V_{GS}^2$$

$$I_{in} \propto V_{in} = e_{in}$$

causes distortion.

DAY-10SWITCHES:-

→ When switch is off - no power dissipation

→ When switch is on - low voltage drop.

ideal switch $+0V-$ closed/on

$V_{in} = 0$ open/off

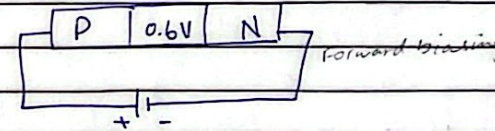
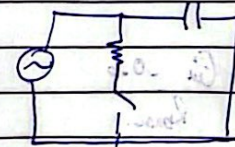
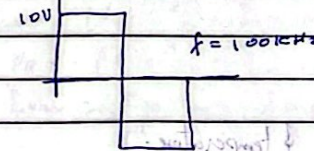
Non-ideal

low volt closed/on

Low side switch:

→ switch circuit common

→ Allowing load to float

Objective:

if we apply high frequency, it will operate on both sides

$$t_{sw} = \frac{1}{f} \text{ signal off}$$

Schottky Diode:

No problem as seen faced in simple diode.
0.3V drop.

Switching Characteristics:

$$I_{base} = \frac{E_{in} - 0.6}{R_{base}}$$

Collector Current:

$$I_c = \beta I_b$$

$$\beta = 20$$

Hard Switching:

Direct on & off

Soft Switching:

Specific current on which switch can be turn on & off.

$$I_c = \frac{E_{supp} - V_{CE(sat)}}{R_{load}}$$

where $0.2 < V_{CE(sat)} < 1.0 \text{ V}$. V_{CE} depends on transistor, I_c & temperature.

$$I_b > \frac{I_c}{\beta}$$

Examp^l.

$$-E_{supp} + V_{load} + V_{CE} = 0.$$

$$-28 + I_c \times 15 + 0.8 = 0.$$

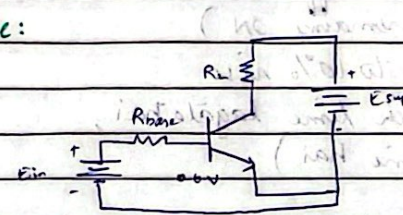
$$I_c \times 15 = 27.2$$

$$I_c = 27.2$$

$$15.$$

DAY-11

Example:



Sol

For the circuit, $E_{sup} = 28 \text{ Vdc}$, $E_{in} = 5 \text{ Vdc}$, $R_{load} = 15 \Omega$

$$V_{CE} = 0.8 \text{ V}$$

$$\beta = 20$$

Calculate the following I_c , P_{load} , I_{base} , R_{base} , P_Q

$$I_c, P_{load}, I_{base}, R_{base}, P_Q$$

$$I_c = \frac{E_{sup} - V_{CE(sat)}}{R_{load}} \Rightarrow 1.81 \text{ A}$$

$$P_{load} = (1.81)^2 (15) = 49.14 \text{ W}$$

$$P_Q = (1.81)(0.8) = 1.45 \text{ W}$$

$$I_{base} > \frac{I_c}{\beta} \Rightarrow I_{base} = 90.5 \text{ mA}$$

$$R_{base} < \frac{E_{sup} - \text{voltage drop}}{I_{base}} = \frac{5 - 0.6}{90.5 \text{ mA}}$$

$$R_{base} = 48.61 \Omega$$

we will use 33Ω

High frequency testing

$$E_{supply} = 12 \text{ V}$$

10 KHz, 50% duty cycle, 5V input

$$\text{When ON, } V_{ce} = 0.849 \text{ V}$$

11-YAG

When input goes low, turns off and $V_{ce} = 12V$

Storage time: (transistor remains ON)

Middle of input edge to 10% rise

(Foran OFF nai hota kuch time lagata hai, wo time shortage time hai)

Rise time: (V_{ce} rises from 10% to 90%)

depends on load resistance and capacitance

Delay time: (to turn transistor ON)

(Jo ON hone mein wo time lagaye ga)

Middle of rising edge to 90% in V_{ce} .

Fall time: (V_{ce} falls from 90% to 10%)

Problems with Darlington Pair:

Darlington pair cannot switch above audio range

Interference (too much annoying)

Solution: R_{in} circuit at 100KHz

Storage time becomes greater than pulse width

Transistor never goes off

In prev. circuit, $f = 10KHz$, produces high pitch w-hine

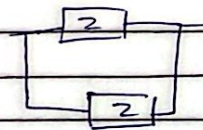
Enhancement mode MOSFET:

can be used as switch

Larger than gate-to-source voltage, larger the channel b/w the source & drain.

$10V < V_{ce} < 20V \rightarrow$ To turn the transistor ON

Question:



$\rightarrow$ voltage change hogi

$\rightarrow$ Power bhi change hogi

DAV-12

Example:

$$I_c = 50mA$$

$$\beta = 100$$

$$I_b = 0.5mA = 500\mu A$$

$$R_b < \frac{V_{in} - V_{BE}}{I_b}$$

$$500\mu A$$

$$= \frac{2.4 - 0.9}{500\mu A}$$

$$= 3K$$

$$R_b = 2.21K\Omega$$

$$I_c = \frac{E_{sup} - V_{ce}}{R_L}$$

$$P_L = \frac{28 - 0.3}{50mA}$$

$$R_L = 554\Omega$$

$$R_{load} = 560\Omega$$

$$V_{BE} = 0.9V$$

$$V_{CE} = 0.3V$$

$$P_L = (I_L)^2 (R) (D)$$

$$= (50mA)^2 (560) (D)$$

$$= (1.4W) (0.5)$$

$$P_L = 0.75W$$

$$P_g = 0.5 \times 50mA \times 0.3$$

$$P_g = 7.5mW$$

Date:.....

Sun Mon Tue Wed Thu Fri Sat

Parasitic capacitance Effect:

Effect on rise & fall time

$$t_{rise} R_c = 2.2 R_L C_p$$

Example.

$$t_{rise} = 2.2(560k)(640p)$$

$$t_{rise} = 788 ns$$

16%

51-VAD

100kHz

$$T = 10\mu s$$

$$D = 50\%$$

$$T = 5\mu s$$

can be ignored

$$(t_{rise} = 2.2(640p)(3.3k))$$

$$(t_{rise} = 4.7\mu s)$$

cannot be ignored

